

Dylan McDermott

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EDUCATION

Iowa State University, Ames, IA	January 2022 - May 2024
<i>Master of Science, Meteorology</i>	GPA: 3.52/4.0
• Thesis: Impacts of U.S. Deforestation on Rainfall from Mesoscale Convective Systems	
Iowa State University, Ames, IA	August 2018 - May 2022
<i>Bachelor of Science, Meteorology</i>	

WORK EXPERIENCE

<i>Weather & Catastrophe Analyst</i>	January 2025-Present
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Property & Liability Resource Bureau, Remote

- Build and maintain production Python and ArcPy systems that ingest and validate NOAA, IEM, radar, observation, warning, and model data; generate daily maps, archives, exports, email content, and ArcGIS updates.
- Process hundreds to thousands of reports, files, and features per day while reconciling late reports and detecting duplicates, missing timestamps, invalid identifiers, incomplete location fields, and silent download failures.
- Develop claims-facing ArcGIS Experience Builder applications for hail, wind, hurricanes, and current weather, including custom REST API integrations for date filtering, layer control, map synchronization, and location-specific reports.
- Evaluate new datasets for scientific defensibility, spatial and temporal resolution, bias, provenance, update behavior, and appropriate interpretation; distinguish observations from modeled and remotely sensed estimates.
- Automated recurring production tasks that previously required hours of manual work while keeping maps, tables, exports, and downstream records synchronized.

<i>Field Sensing Research Associate (Contract)</i>	June 2024-December 2024
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Corteva Agriscience, Johnston, IA

- Co-developed and operated the Smartstick, a wheeled field-sensing platform that collected thousands of timestamped, Trimble GPS-linked canopy and air-temperature measurements per walk across seven research sites.
- Independently built the Python and ArcPy pipeline from raw tables through visual quality control, outlier and GPS filtering, point creation, inward plot buffers, spatial joins, maps, tables, shapefiles, and plot-level summaries.
- Automated approximately one hour of processing per collection, ultimately weeks of processing across the campaign, and applied one reproducible workflow across sites.
- Compared field measurements with drone imagery, LiDAR-derived canopy structure, treatments, and crop-stress response to validate spatial patterns and research findings.
- Co-designed and built a 16-chamber automated soil-gas system, a multi-instrument reference weather station, and tracking dashboards for more than 200 deployed infrared radiometers.

<i>Graduate Research Assistant</i>	May 2022-May 2024
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Iowa State University, Ames, IA

- Designed and ran WRF and Noah-MP experiments at multiple resolutions to quantify how U.S. land-use change affected Midwest rainfall, surface fluxes, moisture transport, and mesoscale convective systems.
- Prepared model inputs by translating CESM/LUMIP land-use data to WRF and Noah-MP grids and evaluating present-day versus 1850 vegetation scenarios.
- Processed and compared ERA5, CESM, WRF, GOES, MODIS, MRMS, radar, and observational datasets using Python, xarray, MATLAB, NumPy, and NetCDF.
- Produced geospatial analyses, derived fields, maps, scientific visualizations, technical writing, and presentations for interdisciplinary research.

<i>Undergraduate Research Assistant</i>	May 2021-August 2021
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Iowa State University, Ames, IA

- Conducted boundary-layer meteorology research by identifying cases, handling large observational and gridded atmospheric datasets, and producing visualizations to interpret physical mechanisms.

SKILLS

• Geospatial data engineering and spatial analysis • Python and ArcPy workflow automation • Raster/vector processing and map production • Data quality control, validation, and reconciliation • Remote sensing, drone imagery, and LiDAR • REST API integration and operational monitoring	• Meteorological and catastrophe analysis • Radar, GOES, MODIS, MRMS, and NWP data • WRF/Noah-MP modeling and simulation • Field sensing and GPS-linked instrumentation • Machine learning and AI-supported analysis • Technical communication and uncertainty analysis
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SOFTWARE

• Python, SQL, MATLAB, R, C++, FORTRAN • ArcGIS Pro, ArcPy, Enterprise/Server, Experience Builder	• NumPy, xarray, OpenCV, Jupyter, NetCDF, AWS • Linux, Windows, Git, GR2Analyst
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PERSONAL PROJECTS

AI Weather Intelligence Platform	2025-Present
• Developing a full-stack platform that integrates 200+ weather and environmental data streams into source-backed historical, current, and forecast analyses. • Builds automated evidence workflows and user-facing geospatial products that translate complex environmental data into maps, plots, reports, and actionable answers. • AWS 10,000 Aldeas Competition Semi-Finalist.	

AWARDS

Esri Special Achievement in GIS (SAG) Award	2025
• Contributed to PLRB catastrophe analytics and ArcGIS applications recognized for innovative and impactful use of GIS technology.	